

**Fourth Semester B. Tech. (Electronics and Communication
Engineering) Examination**

ANALOG AND DIGITAL COMMUNICATION

Time : 3 Hours]

[Max. Marks : 60

Instructions to Candidates :—

- (1) All questions are compulsory.
 - (2) All questions carry marks as indicated against them.
 - (3) Assume suitable data and illustrate answers with neat sketches wherever necessary.
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1.
 - (a) A carrier power 400 W is modulated on depth 75%. Determine total power in DSBFC, DSBSC and SSB. Comment on the results. 5(CO1)
 - (b) A 20 MHz carrier is modulated by 400 Hz modulating signal. The carrier voltage is 5 V and maximum deviation is 10 KHz. Write down standardized mathematical equation and bandwidth for FM and PM waves. 5(CO1)
 2.
 - (a) Illustrate the effect of White Noise in DSBSC system. 5(CO3)
 - (b) What is the need of Pre-Emphasis and De-Emphasis in FM ? Describe in detail Pre-Emphasis and De-Emphasis in FM. 5(CO3)
 3.
 - (a) Derive an expression for SNR in PCM for analog signal input. 5(CO1)
 - (b) Illustrate the importance of non uniform quantization. Explain in detail Adaptive delta modulation. 5(CO1)
 4. Describe in detail MSK Transmitter and Receiver with mathematical expression. Draw the output waveforms for MSK Transmitter, if input is 10010110. 10(CO4)

5. A Convolution Code is described by $X1=[1\ 0\ 1]$; $X2=[1\ 1\ 1]$; Draw the convolution encoder and state diagram. If the received sequence for above encoder is 100111, decode the sequence using Viterbi decoding algorithm. 10(CO5)
6. Derive an expression for Probability of errors for Amplitude Shift keying and Phase shift keying. 10(CO2)

